

# SAITEJA SINGIRIKONDA

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## PERSONAL SUMMARY

Software Engineer with 3+ years of experience designing scalable systems, automating workflows, and leading cross-functional teams. Engineered cloud-native solutions with Java, Python, and cloud technologies (AWS, Azure), with expertise in system design, microservices, and CI/CD pipelines. Specialized in performance optimization, data-driven solutions, and cloud security, ensuring resilient and efficient architectures.

## PROFESSIONAL EXPERIENCE

**Software Engineer** | The George Washington University | *Washington, DC* Aug 2023 – Present

- Engineered a real-time sports analytics pipeline using **Python, Flask, and PostgreSQL**, cutting data processing latency and enabling coaches to assess athlete performance 30% faster through interactive dashboards.
- Designed **RESTful APIs using SQLAlchemy** to automate live game metrics retrieval with **99% accuracy**, improving data accessibility and supporting real-time decision-making for **5+ sports teams** through optimized analytics.
- Built statistical models with **NumPy and Pandas**, improving game strategy by 15% through actionable player insights.
- Enhanced Tableau dashboards by integrating **Matplotlib/Seaborn visualizations**, improving accessibility and reducing review time by **10%**.

**Software Development Engineer** | Wipro Technology | *Bengaluru, India* Dec 2021 – Dec 2022

- Architected a microservices-based test automation framework using **Java, Selenium, and Cucumber**, reducing defect detection time by 70% (from 10 to 3 hours per sprint) and accelerating release cycles for **15+ enterprise applications**.
- Led the implementation of CI/CD pipelines via **Azure DevOps and Jenkins**, automating build, test, and deployment processes to reduce manual effort by **40%** and deployment failures by **30%** across 3 product teams.
- Refined **SQL queries and Flask APIs** through indexing and connection pooling, reducing data retrieval latency from **200ms to 130ms** and enhancing real-time data synchronization for **10K+ daily transactions**.
- Mentored three junior engineers, **leading to a 25% improvement** in testing efficiency and better automation adoption.

**Software Developer** | APPSeCONNECT (formerly Veon Consulting) | *Hyderabad, India* Mar 2020 – Sep 2021

- Directed a 4-member team in migrating over **150K Salesforce records** using REST APIs and SQL, ensuring **98% data accuracy** and minimizing system downtime by **20%** to enhance operational efficiency for enterprise clients.
- Created and deployed API integrations using **XML, JSON, and XSLT** to enhance data flow between Salesforce, Shopify, and NetSuite, increasing operational efficiency by **40%** and reducing manual processing errors.
- Implemented and integrated RESTful APIs using **Flask and Node.js** to replace manual data entry, leading to a **50% reduction** in effort while streamlining **10+ workflows** in cloud environments for improved accuracy.

## PROJECTS

**AWS-Powered CI/CD Pipeline Automation** | *AWS, Terraform, Docker, Kubernetes, Jenkins* Jan 2024 – Apr 2024

Automated end-to-end CI/CD pipelines for containerized applications using Jenkins, Docker, and Kubernetes.

- Provisioned AWS infrastructure using Terraform, improving deployment speed and reducing manual provisioning time by **50%**.
- Optimized **AWS Lambda & S3 for serverless workflows**, improving cloud resource utilization by **40%**.
- Integrated monitoring and logging** using **Prometheus and Grafana**, reducing system downtime by **30%**.

**16-bit Computer System Simulator** | *Java, Eclipse, Pipelining, JUnit* Aug 2023 – Dec 2023

A custom CPU simulator implementing pipelining, caching, and speculative execution for improved instruction processing.

- Developed a pipelined CPU simulator with **branch prediction and cache optimization**, reducing instruction processing time by **20%** and used by **50+ students** in a computer architecture course at GWU.
- Engineered a custom memory management system with **ROM loading and speculative execution**, optimizing resource utilization and enhancing simulation efficiency.

## EDUCATION

**Master's in Computer Science** | The George Washington University | GPA: 3.7 Jan 2023 – Dec 2024

**Bachelor's in Computer Science** | Anurag College of Engineering | GPA: 3.3 Jun 2017 – Aug 2021

## SKILLS

**Programming Languages:** C++, Java, Python, JavaScript, TypeScript, .NET, HTML5.

**Backend & API Development:** REST APIs, Microservices, Flask, .NET Core, Spring Boot.

**Cloud & DevOps:** AWS (EC2, S3, Lambda, DynamoDB, SNS, SQS), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins.

**Databases:** PostgreSQL, DynamoDB, MongoDB, MySQL, NoSQL (Query Optimization, Indexing).

**Frontend Development:** Angular, React, TypeScript, HTML5, CSS.

**Testing & Automation:** Selenium, TestNG, PyTest, Cucumber, Cypress, REST Assured, Postman, JIRA (Defect Tracking).

**Operating Systems:** Linux, Windows, MacOS. | **Soft Skills:** Communication, Teamwork, Leadership, Agile Methodologies.

**Relevant Concepts:** OOP, DSA, Scalable Systems, Agile, Distributed Systems Technical Writing, Agile Methodologies.

**Certifications:** Software Engineering (Elite Certificate) — NPTEL, AWS Academy Graduate – Cloud Foundations, Cloud Web Application Builder, Cloud Security Foundations.